

Your grade: 96%

Your latest: 96% • Your highest: 96% • To pass you need at least 80%. We keep your highest score.

Next item →

coach

Almost there! Let's review your answers and then try again.

Your work showed a good understanding of the primary purpose of the Autograd module, hyperparameters, the role of the torch.save function, and the steps involved in the training loop of a PyTorch model. However, there is room for improvement in understanding the steps involved in building a simple computational graph in PyTorch. Before retrying this assignment, we recommend focusing on this key area:

- **From Tensors to Computational Graphs (101):** In Question 2, your answer missed some key steps involved in building a simple computational graph in PyTorch. Try reviewing [this item](#) to refresh your memory.

Good job on the rest of the assignment!

Retry assignment



1. What is the primary purpose of the Autograd module in PyTorch?

1 / 1 point

- ☐ To create and manipulate tensors
- ☐ To load and preprocess data
- ☒ To automatically calculate gradients
- ☐ To build neural network layers

Correct

Correct! The Autograd module in PyTorch is specifically designed to automatically calculate gradients, which are essential for optimizing neural networks.

2. Which of the following steps are involved in building a simple computational graph in PyTorch?

0.8 / 1 point

- ☐ Initialize weights as zeros
- ☐ Save the model in .pt format
- ☐ Perform operations on tensors
- ☒ Use the backward() function to compute gradients

Correct

Correct! Using the backward() function computes the gradients, which is a crucial step in training neural networks.

☒ Define tensors with requires_grad=True

Correct

Correct! Setting requires_grad=True allows PyTorch to track operations on tensors, which is essential for building the computational graph.

You didn't select all the correct answers

3. Which of the following hyperparameters can impact the performance of a neural network model?

1 / 1 point

☒ Number of layers

Correct

Correct! The number of layers can affect the model's capacity to learn and generalize.

☒ Activation function

Correct

Correct! The choice of activation function can affect the non-linearity and learning capability of the model.

☐ Output data type

☒ Batch size

Correct

Correct! The batch size can affect the stability and efficiency of training.

☒ Learning rate

Correct

Correct! The learning rate is crucial as it determines the step size during gradient descent.

4. What is the role of the torch.save function in PyTorch?

1 / 1 point

- ☐ To train the model
- ☐ To define the model architecture
- ☒ To save the state dictionary of a model for future use
- ☐ To initialize model parameters

Correct

Correct! The torch.save function is used to save the state dictionary of a model so it can be loaded and used later.

5. Which of the following steps are involved in the training loop of a PyTorch model?

1 / 1 point

- ☐ Parameter initialization
- ☒ Forward pass

Correct

Correct! The forward pass involves passing the input data through the model to obtain predictions.

☒ Optimizer step

Correct

Correct! The optimizer step updates the model parameters based on the calculated gradients.

☐ Model evaluation

☒ Backward pass

Correct

Correct! The backward pass involves calculating the gradients of the loss with respect to the model parameters.